



Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie

School of Engineering

Department of Electrical, Electronic and Computer Engineering

Part 1. General Study Guide for all modules presented by the Department of Electrical, Electronic and Computer Engineering

Function head for undergraduate studies: Prof T Hanekom

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ORGANISATIONAL COMPONENT

This guide serves as **Part 1** of all study guides in the Department of Electrical, Electronic and Computer Engineering. The information in this guide is complemented and supported by comprehensive content available on the **EECE Undergraduate ClickUP module**. Part 1 of the study guide will also be referred to as the *general study guide*.

Part 2 of the study guide is module-specific and will be compiled and provided by the lecturer for each module. These study guides will also be referred to as the *module study guides*.

1. GENERAL PREMISE AND EDUCATIONAL APPROACH

The overall objective of the undergraduate programs offered by the Department of Electrical, Electronic and Computer Engineering is to provide world-class education in Electrical, Electronic and Computer Engineering. This will equip graduates with the skills-base to make significant contributions in the engineering industrial and research environments. Since our community, both local and global, is technology-based, competent engineers are essential to the development, support and maintenance of the sophisticated foundations that support society. The Department thus sees its role not only in the technical education of engineers, but also in the development of a professional ethic that guides the conduct of our graduates.

2. HEAD OF UNDERGRADUATE STUDIES

The head of undergraduate studies is responsible for all aspects related to the day-to-day operation and management of undergraduate matters.

2.1. Academic

Prof Tania Hanekom

Email: tania.hanekom@up.ac.za

For appointments, kindly visit the Communication page on the EECE Undergraduate ClickUP module.

2.2. Administrative

Ms Cornel Freislich

Tel: 012 420 3735

Email: cornel.freislich@up.ac.za

Office: Engineering 1, office 15-, University of Pretoria Main Campus

Ms Gawa Pritchard

Tel: 012 420 2775

Email: gawa.pritchard@up.ac.za

Office: Engineering 3, office 7-19, University of Pretoria Main Campus

3. DEPARTMENTAL POLICY ON CONSULTATION AND COMMUNICATION WITH LECTURERS

3.1. Consulting hours

Refer to the module study guides and websites for specific arrangements with regard to consultation hours. Hours for consultation of lecturers, tutors and teaching assistants will be announced at the beginning of the semester, and will also be published on each module web page. Students may consult lecturers, tutors and teaching assistants only during the consulting hours as indicated, or by appointment. This policy also holds before tests and exams. In other words, lecturers, tutors and teaching assistants are only available during their normal consulting hours on the day before a test or examination. This policy aims to encourage students to plan their work and to work continuously.

3.2. Communication with lecturers and other members of staff

Communication with lecturers and other members of staff is regarded as an opportunity to develop students' skills to communicate with a higher level of management. As such, it is expected that students cultivate a professional attitude towards such communication, which must be reflected both in oral and written communication. Kindly note that email messages are considered formal communication and should be approached as such. For guidelines with regard to the style and format of formal communication, refer to the EECE Undergraduate ClickUP module. Lecturers should be addressed by their titles, e.g. Mr Badenhorst and Prof Malherbe.

4. DEPARTMENTAL POLICY ON LECTURES AND CLASS ATTENDANCE

Lectures are presented in a style of co-operative and student-centred learning. Lecture content and structure is based on the prescribed and recommended study material. Note, however, that various problems and examples that are not in the prescribed textbook(s) or study materials may be discussed during lectures.

Attendance of all lectures, including pre-practical and tutorial sessions, is compulsory and register of attendance may be taken on a regular basis. Please note the following:

- If a lecture is missed as a result of a valid reason, such as illness or death in the family, the lecturer should be notified and supporting documents, such as a doctor's certificate, should be submitted to the lecturer if available.
- Students who miss a lecture for invalid reasons may forfeit their right to consultation on the material that was covered in the lecture.
- Students who miss more than three (3) lectures, equivalent to 95% attendance, throughout the course of the semester, for invalid reasons may forfeit their right to any consultation in the module for which the lectures were missed, and may not be eligible for special concessions in the module, e.g. consideration by the department's examination refusal committee for entry into the examination in borderline cases. The final decision regarding the application of these penalties will lie with the lecturer of the specific module and a student will not be allowed to appeal in cases where lecturers decide to strictly apply these penalties.
- Important announcements with respect to laboratory sessions, assignments, tutorials and discussion of test content may be made during lectures. Non-attendance of lectures shall

not be acceptable as an excuse for failure to attend or complete sessions or assignments scheduled during lecture times.

- Progressive evaluation and presence tests: students are expected to prepare for classes. The lecturer may at any time administer a class test on the material covered thus far in the module as well as the material that had to be prepared for that specific lecture. These test marks may contribute to the semester mark according to the details published in the module-specific study guide.
- As a rule no lectures will be cancelled unless special circumstances apply.

5. DEPARTMENTAL APPROACH TO AND POLICY ON HYBRID TEACHING AND LEARNING

Since the beginning of 2016 the university actively started to adopt a hybrid teaching and learning approach to enrich the learning experience of students and exploit the advantages that technology offer to improve the efficiency of teaching strategies. This decision followed on research that showed that a hybrid approach leads to greater levels of student success than models that rely solely on online or face-to-face teaching. The University of Pretoria defines a hybrid approach to teaching and learning as a strategy that is *characterised by different combinations of contact and online delivery. A choice is made by the lecturer to replace some contact time with online time. The module is redesigned from an instructional design perspective to ensure integration and coherence and to achieve the aims of the hybrid model.* These aims can be broadly summarised as (i) to allow students to access a wider array of learning resources, (ii) to exploit the affordances of online technologies, (iii) to develop students as self-regulated learners, (iv) to make individualised learning more achievable, and (v) to enable students to remain in contact with and interact with lecturers, fellow students and resources.

Being a technology-driven entity, the Department EECE realised the potential of technology-supported education early on and initialised the development of innovative technological support for a number of modules at around 2010 and also adopted extensive use of the University's online teaching and learning platform ClickUP in our modules since 2013. As such the Department observes a broader definition of hybrid teaching and learning that includes a wide range of augmentative technology to support teaching and learning. An example of innovative technological support that was developed in and by the Department EECE is the electronic autograding system (AGS). This system has since been expanded to not only grade computer-based assessments, but also hand-written assessments to support large classes. Furthermore, the AGS has been developed into a computerised tutoring system (CTS) to allow 24/7 detailed feedback on tutorial assignments.

The Department of EECE is continuously adjusting the delicate balance among the various available teaching and learning resources to optimize student success while maintaining excellent standards that are on par with the quality of engineering education that is offered by recognised international institutions. With the University's formal adoption of the hybrid approach, the Department will further develop technological support for teaching and learning, which will include a hybrid approach to contact time where some face-to-face activities will be replaced by online activities where appropriate and feasible. This will expand our current portfolio of technology support for teaching and learning. A brief outline of technology supported teaching and learning strategies, of

which an appropriate selection and combination is applied in each module as determined by the department, lecturer and content of the module, is given below. During times of restricted campus access, it may be necessary to rely more heavily on the strategies listed and students must ensure that they are comfortable with the specific technology-supported teaching and learning strategies that are followed in each module.

Students are encouraged to prepare an off-campus study environment where they have access to computing facilities and the internet to enable them to participate in hybrid teaching and learning activities. A dedicated personal or laptop computer with broadband internet access is preferable, but internet cafes are also suitable to allow participation in most online activities. Smart phones and tablet computers offer limited computing functionality and should not be relied on as primary computing facility. A buddy system where students can share off-campus resources is also encouraged since many online activities may be undertaken in small groups.

The hybrid/online approach to teaching and learning that will be followed in the department may, among others, involve any one or more of the following.

- Tutorial lectures: students may be required to complete tutorial assignments at home while some or all face-to-face discussions of the assignments are replaced by online discussions through a discussion platform such as ClickUP's Blackboard Collaborate. Online tutorials will be presented in the tutorial periods as scheduled in the normal timetable unless other arrangements are made.
- Video lectures: suitable content may be presented through pre-recorded video material or live through Blackboard Collaborate. Video lectures will be presented in the lecture periods as scheduled in the normal timetable unless other arrangements are made.
- Video tutorials: step-by-step instruction on selected topics, e.g. the use of laboratory equipment, specific software, preparation for practical assignments, etc. to support content on these topics that were delivered during face-to-face contact sessions.

Other online/technology support that is available in the Department are the following.

- Departmental undergraduate ClickUP platform to support the entire EECE undergraduate student body. This platform also contains study resources such as video tutorials on the use of laboratory equipment.
- EECE undergraduate WhatsApp platform: 079 738 4828
- Electronic grading of tests and examinations.
- Re-deployment of tests and examinations for which electronic grading is available as an online computerised tutoring system (CTS) in selected modules. The system will be rolled out to more modules over time.

6. DEPARTMENTAL POLICY ON ASSIGNMENTS

As a default policy, all assignments that contribute to the semester mark must be submitted to be eligible to gain entry into the examination of a module, i.e. if the module study guide does not specify the rules of assessment for assignments, this policy applies. Arrangements regarding submission of assignments stated in the module study guide take precedence if different from the default policy stated here.

Please note the following.

- If a *hardcopy* of an assignment needs to be handed in, a student needs to make a photocopy of the assignment before handing it in. Hardcopies of assignments must be signed in on a list kept by either the lecturer, assistant lecturer or the secretary. If this procedure is not followed, the assignment shall not count towards the semester mark since no record will be available to verify submission of assignments.
- If a *softcopy* of an assignment needs to be submitted via ClickUP, it is the student's responsibility to ensure that the assignment is submitted on time. Problems with connectivity off-campus will not be an excuse for non-submission or late-submission of assignments. Students need to start the upload process at least one hour before the closing time of the submission to avoid missing the due time as a result of connectivity problems. If a problem is experienced with submission, the assignment needs to be emailed to the lecturer with an explanation of the problem that was experienced BEFORE the closing time of the submission. Note that the ClickUP activity log of a student that submits an assignment via email will be accessed to verify that a submission was attempted in time.

6.1. Standard cover page for all assignments

Group assignments

All group assignments must contain the table and declaration below. Assignments that do not contain these items will not be accepted, i.e. it will be regarded as not submitted. A template for the group assignment submission page is available on the EECE Undergraduate ClickUP page.

Name and Surname	Student number	Signature	% contribution
S. Tudent	12345678		20
S.T. Udent	23456789		35
S.T.U. Dent	34567890		45

By signing this assignment we confirm that we have read and are aware of the University of Pretoria's policy on academic dishonesty and plagiarism and we declare that the work submitted in this assignment is our own as delimited by the mentioned policies. We explicitly declare that no parts of this assignment have been copied from current or previous students' work or any other sources (including the internet), whether copyrighted or not. We understand that we will be subjected to disciplinary actions should it be found that the work we submit here does not comply with the said policies.

For the example above, the final mark for each student will be calculated as follows:

Assume a mark of 8/10 for the assignment.

- S.T.U. Dent gets 8/10
- S.T. Udent gets $8/10 \times 35/45 = 6.2/10$
- S. Tudent gets $8/10 \times 20/45 = 3.6/10$

Students are encouraged to be honest in the specification of the percentage contribution of each group member, since it is another exercise in professional conduct as well as development of self and peer assessment skills. Integrity is an essential part of professionalism. Self and peer assessment are vital skills when working on and managing engineering projects.

Individual assignments

All individual assignments must contain the table and declaration below. Assignments that do not contain these items will not be accepted, i.e. it will be regarded as not submitted. A template for the individual assignment submission page is available on the EECE Undergraduate ClickUP page.

Name and Surname	Student number	Signature
S. Tudent	12345678	

By signing this assignment I confirm that I have read and are aware of the University of Pretoria's policy on academic dishonesty and plagiarism and I declare that the work submitted in this assignment is my own as delimited by the mentioned policies. I explicitly declare that no parts of this assignment have been copied from current or previous students' work or any other sources (including the internet), whether copyrighted or not. I understand that I will be subjected to disciplinary actions should it be found that the work I submit here does not comply with the said policies.

7. DEPARTMENTAL POLICY ON LABORATORY WORK

Please consult the EECE Undergraduate ClickUP module for general information regarding practical sessions, such as the laboratory rules and after-hours access to the labs.

Attendance of all practical sessions is compulsory, i.e. failure to attend a practical session immediately disqualifies a student from entry into the examination of a module. The schedule for the practical sessions and division of groups (if applicable) will be provided by the lecturer.

Please note the following.

- Every module has at least three three-hour laboratory-based practical sessions. Simulation assignments may be used to support and complement laboratory-based practicals, but are mostly not used to replace laboratory-based practicals – a few exceptions apply and are delineated in the specific study guides of those modules. This is the minimum requirement and the number of practical assignments may be increased depending on the nature of the module.
- Group or team work is a necessary component of the engineering skill set that graduates need to cope in industry. Because of limited resources, most practicals in the Department have to be completed in teams or groups. If resources allow, a student may choose to complete practical assignments on his/her own. No accommodation in terms of the scope or due date for practical assignments will be made for students who wish to work individually.

Speak to your lecturer if you would like to work individually. However, please note that the decision of whether this will be allowed lies with the lecturer.

- If a student or group attends a practical session unprepared, it will be regarded as non-attendance of the practical session, i.e. immediate disqualification from exam entrance in the module. The default minimum requirement for practical attendance is that half of the subminimum requirements for a practical assignment have been met. Refer to the module study guide for specific details about pass requirements for the practical section of each module.
- Keeping a laboratory notebook ("lab book") as described in the practical guide is compulsory. The lab book may be assessed as part of the mark for each practical. Notes on keeping a laboratory notebook are given on the EECE Undergraduate ClickUP page.
- **Practical reports form part of the assessment criteria for laboratory work and submission of all practical reports, completed to a satisfactory level, is compulsory.** Please refer to the EECE Undergraduate ClickUP module for information on compiling reports.
- **The subminimum for the practical component of a module is 50%.** This may be either 50% on average or 50% for each practical as specified in the module specific study guide.

8. DEPARTMENTAL POLICY ON ASSESSMENT

Also see the examination regulations in the Year Books of the Faculty of Engineering, Built Environment and Information Technology (Part 1: Engineering).

7.1. Tests and examinations

Two semester tests are written in each module during the scheduled test weeks. They count towards the semester mark as specified in the module study guide. Any number of class tests may also be written and may count towards the semester mark.

Please note that all tests that count towards the semester mark must be written to be eligible to gain entry into the examination for a module.

Tests and examinations may be either open or closed book.

- Closed book tests require that the student has no access to study materials during the test or examination.
- Open book tests allow students to have free or limited access to study materials.
 - If *limited access* to study materials is allowed, all allowed materials must be standardised, e.g. prescribed textbooks and/or other materials, e.g. datasheets, lecturer-compiled equation sheets, etc. None of these materials may contain any personalised notes. These materials may be inspected at any time during a test or examination. Students may not share these materials during tests or examinations.
 - If *free access* to study materials is allowed, any materials may be used during the test or examination. This mode of examination also allows for lecturers to have students compile a summary of any materials that they wish to use in a test or an examination. No control over the content of such a summary is exerted. Students may not share these materials during tests or examinations.

7.2. Entrance into the examination of a module

Two subminima are by default required for entry into the examinations for modules in the Department of Electrical, Electronic and Computer Engineering unless otherwise stated in the module study guide:

- Team/group work subminimum of 40%
- Individual work subminimum of 40%

Please refer to the module study guides for a detailed breakdown of semester mark composition.

7.3. Composition of the final mark

The default composition for the final mark is 50% semester mark and 50% examination mark. However, the breakdown as given in the specific study guide for a module takes preference if different from the preceding breakdown.

8. DEPARTMENTAL RESOURCES

A complete list of Departmental resources is available on the EECE Undergraduate ClickUP module.

9. GENERAL

9.1. Absence from formal evaluation opportunities and practical sessions

Absence from tests, practical sessions and assignments. In the event of absence from tests, practical sessions and when compulsory assignments are not handed in, an official and valid statement (e.g. a medical certificate) has to be submitted to Ms. C Freislich (Eng 1, office 13-20) within **three** days of the date of the test. Overdue assignments will not be graded.

Absence from examinations. In the event of absence from an examination an official and valid statement (e.g. a medical certificate) has to be submitted to Engineering administration within **three** days of the date of the examination.

9.2. Grievance procedures

Kindly refer to the procedures outlined under Communication on the EECE Undergraduate ClickUP module.

9.3. Academic dishonesty and plagiarism

Kindly refer to:

- the Plagiarism page on the University of Pretoria library site: <http://www.library.up.ac.za/plagiarism/index.htm>.
- the Disciplinary Code for Students as published on the ClickUP platform.
- the procedures outlined under Academic dishonesty on the EECE Undergraduate ClickUP module.

Students are encouraged to discuss work with one another. However, each student should hand in his/her **own** work for assignments. Plagiarism, which also includes copying the work of another student during tests and exams and copying from the Internet, can lead to expulsion from the University.

Even if another student gives you permission to use his/her assignments or research to hand in as you own, you are not allowed to do it. It is a form of plagiarism. You are also not allowed to let anybody copy your work with the intention of passing it off as his/her own work. Speak to your lecturer if you are uncertain about what is required.

According to the University's regulations a lecturer is obliged to charge guilty students with dishonesty that may lead to suspension.

The statement below regarding the originality of your work must be appended to ALL written work submitted for evaluation in this department:

I, name and surname, hereby confirm that I have read and are aware of the University of Pretoria's policy on academic dishonesty and plagiarism and I declare that the work herewith is completely my own as delimited by the mentioned policies. I explicitly declare that no parts of this assignment have been copied from current or previous students' work or any other sources (including the internet), whether copyrighted or not. I understand that I will be subjected to disciplinary actions should it be found that the work I submit here does not comply with the said policies.

STUDY COMPONENT

1. GENERAL DEGREE PROGRAM OBJECTIVES

The general objective of the teaching and learning strategy implemented by the Department is to develop students' skills to produce graduates that comply with the requirements established for professional engineers through the regulatory body ECSA. The variety and scope of modules that are presented aim to develop a range of engineering skills covering basic sciences such as physics and mathematics to more advanced specialist concepts such as control systems, telecommunications, signal processing and artificial intelligence. The content of the degree programs also aims to equip students to function in a multidisciplinary environment where they need to interface with other engineering and science disciplines.

2. ECSA

The Engineering Council of South Africa (ECSA) is a statutory body established in terms of the Engineering Profession Act (EPA), 46 of 2000 whose role is to regulate the engineering profession. It does this through accreditation of engineering programmes, registration of engineering professionals and regulation of the practice of registered persons.

All three degree programs offered by the Department of Electrical, Electronic and Computer Engineering are accredited by ECSA. In ECSA's *Background to Accreditation of Engineering Education Programmes* document (document E-01-P available from www.ecsa.co.za) the purpose of the accreditation system is stated as follows.

"The accreditation system assures the public, students, employers, funders and other stakeholders that:

- a) The programme fulfils its key purpose, namely to provide the graduate with the educational foundation for engineering in a stated role at the professional level; and*
- b) The teaching, learning and assessment processes is effective."*

For more information on ECSA, please visit the website at www.ecsa.co.za.

2.1. Knowledge Areas

ECSA prescribes a credit allocation according to knowledge areas that needs to adhere to the values given in Table 1 of the document [Qualification Standard for Bachelor of Science in Engineering \(BSc\(Eng\)\)/Bachelors of Engineering \(BEng\): NQF Level 8, Document E-02-PE, Rev-4, 31 July 2014](#), available from the ECSA website. The table is given below.

Knowledge area	Minimum Credits
Mathematical sciences	56
Natural sciences	56
Engineering sciences	180
Design and synthesis	72
Complementary studies	56
Subtotal	420
For reallocation	≥140
Total credits	≥560

The description of the ECSA exit level outcomes below is an excerpt from the document [Background to Accreditation of Engineering Education Programmes, Document E-01-P, Rev-3, 31 July 2014](#), available from the ECSA website.

Natural sciences (formerly basic sciences): physics (including mechanics), chemistry, earth sciences and the biological sciences which focus on understanding the physical world, as applicable in each engineering disciplinary context.

Complementary studies: cover those disciplines outside of engineering sciences, natural sciences and mathematics which are relevant to the practice of engineering including but not limited to engineering economics, management, the impact of technology on society, effective communication, and the humanities, social sciences or other areas that support an understanding of the world in which engineering is practised.

Computing and information technologies: encompasses the use of computers, networking and software to support engineering activity and as an engineering activity in itself as appropriate to the discipline.

Engineering design and synthesis: is the systematic process of conceiving and developing materials, components, systems and processes to serve useful purposes. Design may be procedural, creative or open-ended and requires application of engineering sciences, working under constraints, and taking into account economic, health and safety, social and environmental factors, codes of practice and applicable laws.

Engineering sciences: have roots in the mathematical and physical sciences, and where applicable, in other natural sciences but extend knowledge and develop models and methods in order to lead to engineering applications and solve engineering problems.

Mathematical sciences: an umbrella term embracing the techniques of mathematics, numerical analysis, statistics and aspects of computer science cast in an appropriate mathematical formalism.

The coverage of knowledge areas among the modules in each degree program are given in Appendix A. Note that the coverage allocation is based on the general approach in each module, i.e. it does not mean that none of the other knowledge areas are touched upon in a module where they are not indicated as the primary knowledge area(s).

2.2. ECSA Exit Level Outcomes (ELOs)

The description of the ECSA exit level outcomes below is an excerpt from the document [Qualification Standard for Bachelor of Science in Engineering \(BSc\(Eng\)\)/Bachelors of Engineering \(BEng\): NQF Level 8, Document E-02-PE, Rev-4, 31 July 2014](#), available from the ECSA website.

Each of the modules in the undergraduate programs in the Department addresses one or more of these outcomes with the purpose of complying with the quality of engineering education that is required to practice as a professional engineer.

The exit level outcome descriptions are copied below for quick reference. Kindly consult document E-02-PE above for a detailed description of ECSA Outcome levels.

Exit-level Outcome 1: Problem solving

Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.

Exit-level Outcome 2: Application of scientific and engineering knowledge

Apply knowledge of mathematics, natural sciences, engineering fundamentals and an engineering speciality to solve complex engineering problems.

Exit-level Outcome 3: Engineering Design

Perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works, products or processes.

Exit-level Outcome 4: Investigations, experiments and data analysis

Demonstrate competence to design and conduct investigations and experiments.

Exit-level Outcome 5: Engineering methods, skills and tools, including information technology

Demonstrate competence to use appropriate engineering methods, skills and tools, including those based on information technology.

Exit-level Outcome 6: Professional and technical communication

Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large.

Exit-level Outcome 7: Sustainability and impact of engineering activity

Demonstrate critical awareness of the sustainability and impact of engineering activity on the social, industrial and physical environment.

Exit-level Outcome 8: Individual, team and multidisciplinary working

Demonstrate competence to work effectively as an individual, in teams and in multidisciplinary environments.

Exit-level Outcome 9: Independent Learning Ability

Demonstrate competence to engage in independent learning through well-developed learning skills.

Exit-level Outcome 10: Engineering Professionalism

Demonstrate critical awareness of the need to act professionally and ethically and to exercise judgment and take responsibility within own limits of competence.

Exit-level Outcome 11: Engineering management

Demonstrate knowledge and understanding of engineering management principles and economic decision-making.

The articulation of the outcomes within the three degree programs presented by the Department EECE are given in Appendix A. Three levels of articulation are indicated: ✓ indicates introductory level, ✓✓ indicates intermediate level and ✓✓✓ indicates exit level.

3. COGNITIVE LEVEL OF ASSESSMENT

The criteria of assessment are a list of specific skills to be mastered by the student in order to achieve the learning outcomes of the syllabus theme. During assessment (tests and the examination), students will be evaluated in terms of these criteria.

The statements used to define the criteria of assessment are classified in terms of a series of lower- to higher-order thinking skills (cognitive domains), in accordance with Bloom's Taxonomy of Educational Objectives (Bloom BS and Krathwohl DR, Taxonomy of educational objectives. Handbook 1. Cognitive domain, Addison-Wesley, 1984). A diagram that shows the hierarchical progression of these domains is shown in Figure 1.

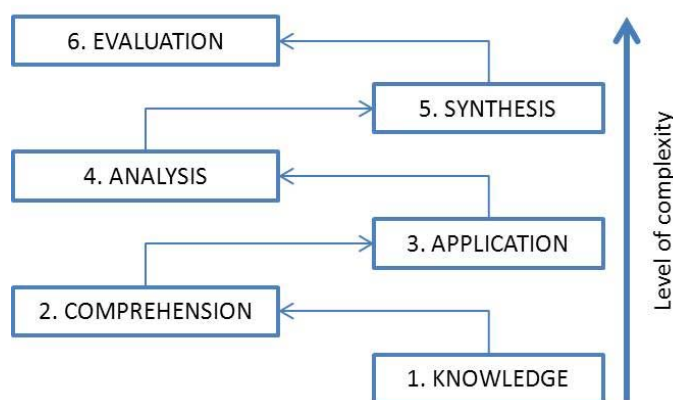


Figure 1. Cognitive levels of complexity

The characterization of the cognitive domains is given in the table below.

Cognitive Domain	Definition	Typical Action Verbs
1. Knowledge	Remembering previously learned information.	Arrange, define, describe, identify, label, list, match, name, outline

2. Comprehension	Understanding the meaning of information.	Classify, discuss, estimate, explain, give example(s), identify, predict, report, review, select, summarise, interpret, "in your own words"
3. Application	Using the information appropriately in different situations	Apply, calculate, demonstrate, illustrate, interpret, modify, predict, prepare, produce, solve, use, manipulate, put into practice
4. Analysis	Breaking down the information into the component parts and seeing the relationships.	Analyse, appraise, calculate, compare, criticise, derive, differentiate, choose, distinguish, examine, subdivide, organise, deduce
5. Synthesis	Putting the component parts together to form new products and ideas.	Assemble, compose, construct, create, design, determine, develop, devise, formulate, propose, synthesize, plan, discuss, support
6. Evaluation	Making judgments of an idea, theory, opinion, etc., based on criteria.	Appraise, assess, compare, conclude, defend, determine, evaluate, judge, justify, optimise, predict, criticise

In any module there are two levels of complexity that are being addressed: the cognitive level of complexity and a general level of difficulty that relates to the complexity of performing, for example, the required mathematics, the number of subsequent steps that are involved or the level of integration of different concepts in the process of solving a problem. Figure 2 shows a graphical representation of this proficiency domain idea.

Up to the second year the content of your modules as well as the way in which they are/were presented mainly require that you work on levels 1 and 2 (knowledge and comprehension) in the cognitive domain and perhaps start touching on level 3 (application). This is roughly indicated by the green domain in figure 2. If your preparation for these junior years results in a distinction mark, it shows that you have mastered the lower cognitive levels to address problems of moderate to difficult nature and on an application level, problems of trivial to easy nature. A mark between 50 and 75% suggests that you demonstrate proficiency in the same cognitive domains, but at a level of easy to moderate scope. A mark below 50% undeniably indicate that although you might be able to address trivial to easy problems in the lower three cognitive domains, you still need to develop the skills that are necessary to address problems at a more difficult level.

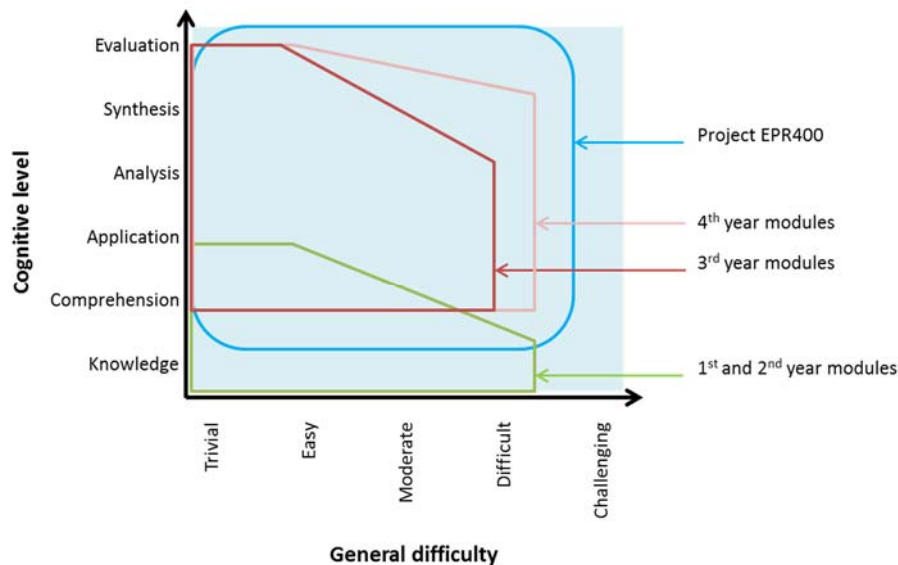


Figure 2. Proficiency domain with focus areas of different study years indicated.

Engineering problems in industry cover the entire range of cognitive complexity and general difficulty as indicated by the shaded area in Figure 2. The objective in our undergraduate modules is thus to broaden the proficiency domain from that indicated for the junior years to a domain that would allow a student to effectively function in industry or academia. The final year project typically covers the proficiency range shown in blue in Figure 2. The preceding modules must deliver students to the final year project EPR400 and ultimately to industry that have the skills required to function as professional engineers.

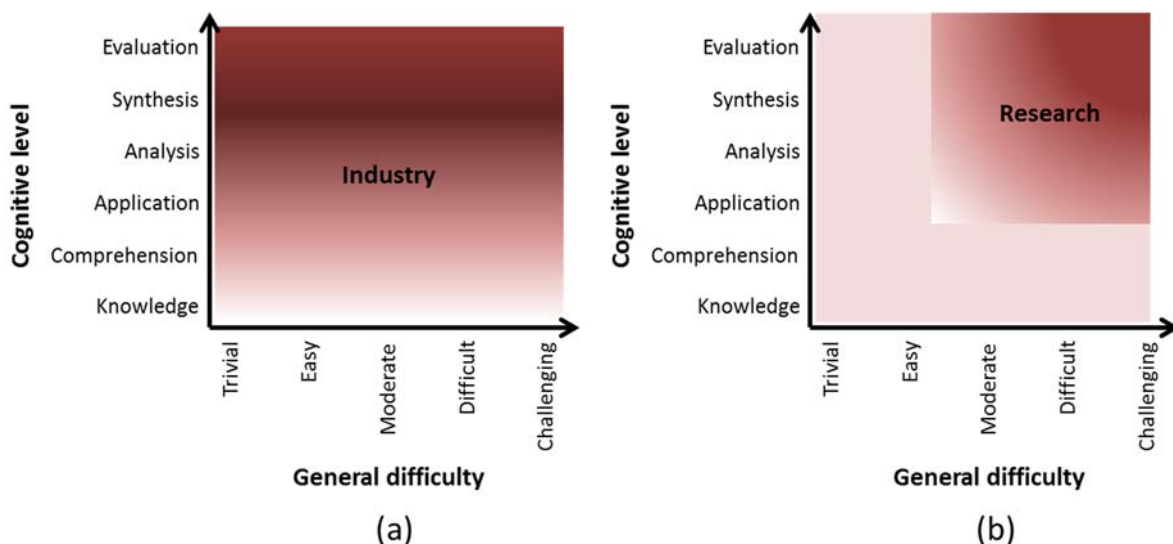


Figure 3. Typical proficiency domains in at which engineers operate in (a) industry and (b) research.

So how far do I really need to develop my skills? Figure 3 roughly indicates the typical domains where engineers in industry (Figure 3a) and engineers in academia (Figure 3b) need to function. As a student you must thus aim to develop your skills to be able to function over the entire proficiency domain with a focus towards the higher cognitive levels of complexity. Research typically tends to

be at the more difficult side of the domain. Industrial applications tend to cover the entire difficulty range. For any engineering application (industrial or research) knowledge and comprehension and even to an extent application skills are assumed to be in place to support the approach to the problem from an analysis-synthesis-evaluation viewpoint.

This concludes the general study guide. Kindly refer to the module study guides for specific content related to each module.

Appendix A: Articulation of modules with knowledge areas and exit level outcomes.

Summary of modules offered and outcomes assessed: Electrical (E), Electronic (Z) and Computer (R) Engineering

[illegible]